

STEM BIO-AI Evidence-Surface Scan v1.6.0

Repository: ClawBio/ClawBio | Commit: c2a73bd621ed | Branch: main | Audit Date: 2026-05-06 | Mode: LOCAL_ANALYSIS

67 / 100

Final Score

T2 Caution

Use Scope:

Research reference and supervised non-clinical technical review only.

Weighted model: Stage 1 x 0.40 + Stage 2R x 0.20 + Stage 3 x 0.40 - Risk Penalty = 67

Stage 1
README Evidence

60 / 100

Stage 2R
Repo-Local Consistency

70 / 100

Stage 3
Code / Bio Responsibility

72 / 100

Stage 4
Replication Evidence

55 / 100

Code Integrity

PASS

C1 Credentials

No direct credential patterns detected by local CLI scan.

WARN

C2 Dependency Pinning

requirements.txt:1 biopython>=1.82

PASS

C3 Deprecated Paths

No deprecated patient-adjacent metadata patterns detected.

WARN

C4 Exception Handling

bot/roboterri.py:133 except OSError:

Remediation Targets

- C2_dependency_pinning: WARN
- C4_exception_handling_clinical_adjacent_paths: WARN

Positive Evidence

- CI workflow files were detected.
- Documentation files were detected.

Bio Deterministic Diagnostics

SMILES Surface Integrity

not_detected=1

SMILES RDKit Validation

not_applicable=1

SMILES Parser Guard

not_detected=1

Silent Mock Fallback

not_detected=1

Traceability Manifest Surface

not_detected=1

Bio Subprocess Run Trace

not_detected=1

Regulatory basis note

Aligned to current official source classes as of May 2026: EU AI Act (Regulation (EU) 2024/1689), FDA QMSR, FDA AI-enabled device guidance themes, and IMDRF SaMD/GMLP frameworks.

This is a traceability aid, not a compliance or clearance determination.

Traceability summary: Structural signals partially align with traceability scaffolding. This remains a pre-audit traceability aid, not a compliance determination.

Stage 1 — README Evidence Signal | Weight: 0.40

60 / 100
S1 Score

Check	Points	Evidence / Finding
Baseline	+60	Non-nascent README evidence baseline.
README present	-20	No root README detected.
R2: Regulatory Framework	-10	CA-DIRECT surface lacks regulatory or governance framework language.
R3: Clinical Boundary	+10	Explicit non-clinical or non-diagnostic boundary detected.
R4: Bias / Subgroup Boundary	+10	Demographic, subgroup, fairness, bias, or validation-cohort language detected.
R5: Reproducibility Provisions	+10	Reproducibility, replication, seed, lockfile, or checksum language detected.

Calculation: 60 plus Stage 1 evidence additions/deductions = 60

• Clinical-Adjacent: **YES** (CA-DIRECT) • Explicit Disclaimer: **PRESENT** • T0 Hard Floor: **Clear**

Stage 2R — Repo-Local Consistency | Weight: 0.20

70 / 100
S2R Score

Check	Points	Evidence / Finding
Baseline	+60	Non-nascent local repository baseline. — Every repository that is not nascent starts at 60. This baseline accounts for basic structural maturity.
R2R-3: README / Test-CI Alignment	+10	Test/CI surfaces are present and locally consistent. — Test and CI surfaces are present and reference the same domain as the README.

60 plus local consistency additions/deductions = 70 **Mixed Local Consistency**

Stage 3 — Code & Bio Responsibility | Weight: 0.40

72 / 100

S3 Score

Engineering Accountability (T-series)

Check	Points	Evidence / Finding
T1: CI/CD Workflow	15 / 15	Workflow files detected. — CI/CD workflows (GitHub Actions, GitLab CI, CircleCI) verify that commits do not silently break the pipeline. Full credit (15) requires workflow...
T2: Domain-Specific Tests	0 / 15	No tests detected. — Domain-specific tests verify biological outputs — e.g., sequencing pipeline correctness, variant call validation, or genomic data integrity. Full credit...
T3: Changelog & Release Hygiene	15 / 15	CHANGELOG.md contains bug-fix, patch, or security entries. — A CHANGELOG tracks which version fixed which defect — essential for regulatory traceability and reproducibility...

Biological Integrity (B-series)

Check	Points	Evidence / Finding
B1: Data Provenance Controls	15 / 15	Dependency manifest detected with data source, IRB, or dataset citation language. — Dependency manifests (requirements.txt, pyproject.toml, environment.yml) establish...
B2: Bias / Limitations Documentation	8 / 15	Bias/limitations language detected; no quantitative measurement evidence found. — Documentation of algorithmic bias, limitations, or model boundary conditions. Score 8 for...
B3: COI & Funding Disclosure	5 / 5	COI, funding, sponsor, or acknowledgement language detected. — Conflict of interest and funding disclosure in README or FUNDING.md. Required for institutional review context...

Stage 3 Gap Analysis — Path to Next Tier

- **T-series (engineering) attained:** 30 / 45 **B-series (bio integrity) attained:** 28 / 35
- **Local CLI scan maximum:** 55 / 100 (T1+T2+T3 max 15 each; B1 max 10; B2/B3 require manual review)
- **Gap to T3 (final score >= 70):** 3 points needed across all stages
- **Gap to T4 (final score >= 85):** 18 points needed across all stages
- **B2 Bias/Limitations:** Not detectable — requires manual audit of README, model card, or supplementary documentation for validation boundaries and algorithmic limitations
- **B3 COI/Funding:** Not detectable — requires inspection of README or FUNDING.md for conflict of interest and funding source disclosure

Code Integrity — Deep Analysis

Check	Points	Evidence / Finding
C1: Hardcoded Credentials	PASS	No direct credential patterns detected by local CLI scan. Scan: Scans for AWS access keys (AKIA*), OpenAI keys (sk-*), GitHub tokens (ghp_*), and api_key = '...' patterns...
C2: Dependency Pinning	WARN	requirements.txt:1 biopython>=1.82 Scan: Checks whether requirements.txt / pyproject.toml / environment.yml use exact version pins (==, sha256 hash) or loose ranges (>=...
C3: Deprecated Patient Paths	PASS	No deprecated patient-adjacent metadata patterns detected. Scan: Scans deprecated/ directories for patient metadata patterns: patient_id, patient_age, patient_sex...
C4: Fail-Open Exceptions	WARN	bot/roboterri.py:133 except OSError: Scan: Detects fail-open exception patterns: 'except Exception: pass' or 'except: return True' in code — these silently ignore errors...

Remediation Guidance

[WARN] C2: Dependency Pinning:
→ Pin all dependencies to exact versions (== for pip, hash-pinning for conda). Run pip-audit or safety regularly. Consider pip-compile for reproducible lock files. Unpinned ranges in clinical-adjacent pipelines create silent regression risk.

[WARN] C4: Fail-Open Exceptions:
→ Replace broad 'except Exception: pass' or 'except: return True' patterns with specific error types and explicit failure logging. In clinical-adjacent code paths, any silent failure is a patient safety risk. Fail closed, not open.

Classification & Repository Analysis

Check	Points	Evidence / Finding
Clinical Adjacent	YES	Severity: CA-DIRECT. Triggered by BIO/CLINICAL_OUTPUT term regex match across README, docs, and code.
T0 Hard Floor	Clear	No T0_HARD_FLOOR condition detected.
Explicit Disclaimer	PRESENT	Regex: 'not for clinical not for diagnostic research use only not medical advice' in README + docs surface.
Files Scanned	498	Total files indexed by recursive walk. Text files only for content analysis; binary files counted but not read.
Execution Mode	LOCAL_ANALYSIS	No LLM calls. No network access. No runtime execution. Deterministic regex + file-system scan only.

File Integrity (SHA-256)

File	SHA-256 Hash
requirements.txt	157F12EF2DB28FDA7458F7E154821B8A1DFE2DDF2E6A82B4B389C416C5C112E6

Priority Improvement Roadmap

Priority 1: C2_dependency_pinning: WARN

→ Pin all production dependencies to exact versions (== for pip). Add pip-audit or safety to CI pipeline for vulnerability scanning. Consider pip-compile for deterministic lock files.

Priority 2: C4_exception_handling_clinical_adjacent_paths: WARN

→ Replace broad exception handlers with specific error types and explicit logging. In any clinical-adjacent code path: fail closed, not open. Never silently return True or pass on exception.

Positive Evidence Summary

- CI workflow files were detected.
- Documentation files were detected.

Method Boundary

Deterministic local CLI scan. No LLM, network, or runtime test execution is required.

Scope boundary: Runtime behavior, model output correctness, dynamic validation, wet-lab reproducibility, and clinical validation are outside the scope of this local CLI scan. This report assesses structural signals only.

Report Metadata

Field	Value
Schema Version	stem-ai-local-cli-result-v1.6
STEM BIO-AI Version	1.6.0
Generated (local date)	2026-05-06
Report Validity	180 days from audit date
Execution Mode	LOCAL_ANALYSIS
Repository	ClawBio/ClawBio
Remote URL	https://github.com/ClawBio/ClawBio
Branch	main
Commit (HEAD)	c2a73bd621edab1db823315007abdc5649af9b1f
Files Scanned	498
Final Score / Tier	67 / 100 — T2 Caution